

VISHING CHALLENGE · PRESENTATION ATTACK DETECTION

Deepfake Voice Detection for Anti-Vishing Systems

A real-time, generalizable Presentation Attack Detection architecture
that runs on the callee's phone and feeds a multi-layer phishing defense

Solution Proposal · 2-Day Sprint · Panel Review

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Voice cloning has outpaced voice defense



Cloning is now trivial

Modern TTS and voice-conversion models build a convincing copy of a voice from a few minutes of public audio, so impersonation no longer takes skill.



Static detectors collapse on new generators

Detectors trained on known synthesis methods do not carry over to generators they have not seen (Li & Chen, 2024).



The stakes sit with the person answering

A cloned voice asking for an urgent transfer reaches the victim directly, with nothing behind it, a single miss is costly.

THE EVIDENCE

How far it falls

TRAINED ON ASVSPOOF 2019, TESTED ON:

its own test set



another corpus

In-the-Wild



2026 cloned voices

over a real VoIP call



COIN FLIP



same test, retrained

on modern voices



equal error rate, lower is better



A 36-point gap between the benchmark and anything else. **Closing that gap is what we are building.**

DIAGNOSIS

It learns the dataset, not the voice

Before choosing an architecture we asked what the failing detectors are actually keying on. The answer explains the collapse, and it tells us what to remove during training.

SILENCE ALONE

85%

accuracy from a classifier given only the length of the opening silence

Muller et al.

ATTACK ON SILENCE ONLY

82.2%

attack success from changing only the background noise and the mute segments, never the voice

SiFDetectCracker, ACM MM 2023

WHERE IT LOOKS

0.5-0.6 kHz

SHAP analysis: the model attends to the gaps between words and one narrow low band

arXiv:2110.03309

The detector wins on what belongs to the corpus, not to the speech. **Every one of those cues can be taken away in training.**

What the panel is asking us to solve



Objective

Develop and test a robust, generalized deep learning model for Presentation Attack Detection against modern audio deepfake and synthetic voice-spoofing techniques, improving voice authentication security while reducing multi-layer phishing risk.

Success means accuracy that holds up on generators it has never seen, not just the ones it trained on.



Real-time

Low enough latency for live-call integration



Multi-layer ready

Feeds an adaptive phishing-defense stack



2-day sprint

Build, test, and prep an expert panel review



20-min readout

Findings + proposed solution to the panel

The core bet: foundation models, not detectors from scratch

Benchmarks across many generators keep showing the same thing. A detector trained from scratch learns the synthesis methods it saw. A large pretrained speech model holds up better, because it learned what real speech is before it ever saw a fake.

Traditional CNN / RNN Detectors

- ✗ Trained end-to-end on spectrograms from known vocoders
- ✗ Strong on in-distribution benchmarks
- ✗ Accuracy drops sharply on unseen generator families
- ✗ Little transfer to a new corpus or channel

SSL Foundation Models (Wav2Vec2 / WavLM)

- ✓ Large-scale pretraining gives richer, more general speech representations
- ✓ Demonstrated cross-generator generalization in benchmark studies
- ✓ Holds up under attack: 11.6% EER where a CNN detector goes to 61.81%
- ✓ Supports few-shot adaptation for high-value protected identities

Detection pipeline, end to end



The trust score is thresholded for a real-time block or allow, then passed on as one signal into the multi-layer defense on the next slide.

Making it real-time, not just accurate



Streaming inference

Overlapping 1 to 2 second windows, with the trust score updating as the call runs rather than at the end.



Model compression

Quantize/distill the SSL backbone (INT8, distilled student) so it keeps up with the live stream on a phone.



Degraded-channel robustness

We measured the official baseline per codec: G.711 costs little (6.3%), but GSM hits 10.7% and uncontrolled PSTN transcoding 16.7%. Compression severity, not bandwidth, is what hurts.



On-device triage

Fast handcrafted-feature triage on the device for an instant first pass; the heavier backbone confirms right after.

One signal in a multi-layer phishing defense



Layer 1: Voice PAD

Real vs. synthetic score per call segment (this project's deliverable)



Layer 2: Behavioral & Context

Caller-ID spoofing signals, call metadata anomalies, urgency / funds-transfer request patterns typical of vishing scripts



Layer 3: Fusion & Adaptive Decision

Combines PAD score with contextual risk; borderline voice + high-risk context triggers step-up auth, high-confidence spoof triggers an immediate block

Grounded in a multi-layer adaptive framework for phishing-attack detection (Ige, Kiekintveld & Piplai, 2024).

Proving generalization, not just accuracy



EER, min t-DCF, per call-minute

Standard PAD metrics on held-out splits. And a streaming detector is judged per call, not per window: at 5% per-window error a 60-second call false-alarms 79% of the time.



Leave-one-generator-out

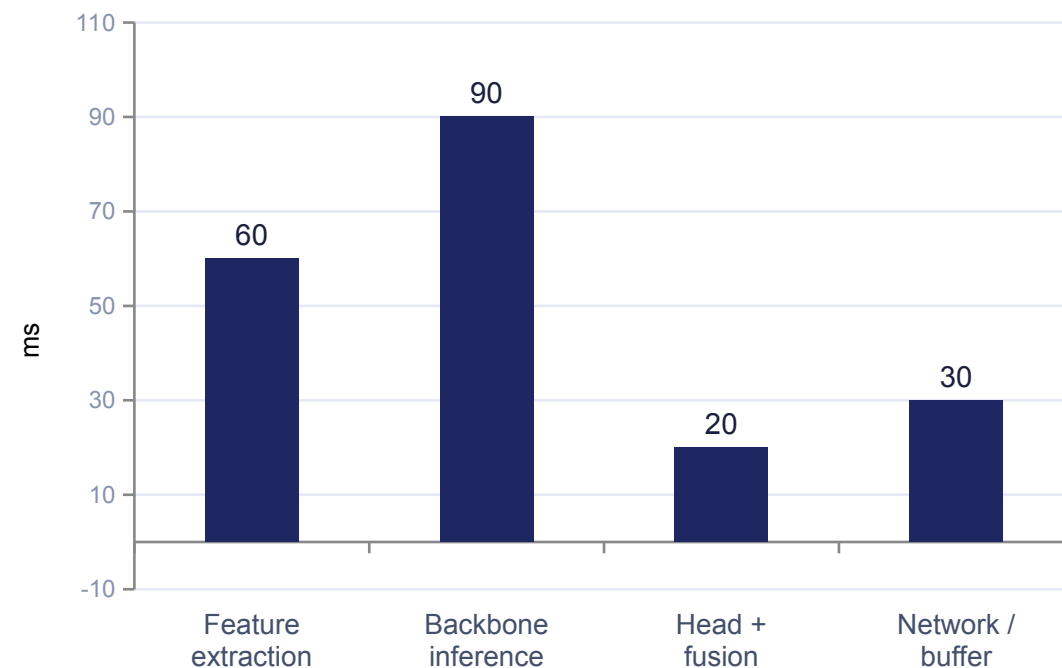
Whole vocoder and TTS families are kept out of training, so cross-generator performance is measured directly.



Reported with and without silence

The corpus ships a VAD-trimmed subset of 16,926 trials. Every EER is reported twice, so no number of ours leans on the silence artifact.

WHERE THE TIME GOES · MS PER 2-SECOND CHUNK



The alarm is read by a person, so the real budget is seconds. Per chunk we only have to keep up with the stream, which is what compression on slide 7 buys us.

From proposal to a working detector

1

2-DAY SPRINT

Literature & proposal

Survey the generalization failure, fix the evaluation protocol, and commit to an architecture. This deck.

2

WEEKS 2-4

Baseline

Reproduce published clean-condition EER and validate against the official challenge baseline scores.

3

WEEKS 5-9

Generalization

Degraded-channel and silence augmentation, foundation-model front-end, leave-one-generator-out. The core work.

4

WEEKS 10-12

Real-time & write-up

Streaming inference, compression, latency on device, then the report and defence.

Limitations & next steps



Known limitations

- A short sprint limits how much training data we can cover compared with production scale.
- Foundation-model backbones cost more compute per stream than a small CNN.
- No dataset fully reproduces a live, adversarial vishing call.



Next steps beyond the sprint

- ✓ Expand few-shot enrollment for high-value protected identities
- ✓ Field-test on real degraded-channel call recordings
- ✓ Formalize the fusion/decision policy with the phishing-defense team

Questions for the panel

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Thank you.

